Surface complexation - the COMPLEX keyword

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Dec 12, 2025

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1 Background

Surface reactions are implemented based on standard Poisson-Boltzmann / Gouy-Chapman theory; see [8, 5, 7, 6, 9]. The following surface complex reactions are included in the database [2, 3] for chalk (see source file `ChemDatabaseHKF_surface.cpp`)

$$>\mathrm{CaCO_3^-} \rightleftharpoons >\mathrm{CaH_2O^+} + \mathrm{HCO_3^-} - \mathrm{H^+} - \mathrm{H_2O}\,, \tag{1}$$

$$>\mathrm{CaOH^0} \rightleftharpoons >\mathrm{CaH_2O^+} - \mathrm{H^+}\,, \tag{2}$$

$$>\mathrm{CaHCO_3^0} \rightleftharpoons >\mathrm{CaH_2O^+} + \mathrm{HCO_3^-} - \mathrm{H_2O}\,, \tag{3}$$

$$>\mathrm{CaSO_4^-} \rightleftharpoons >\mathrm{CaH_2O^+} + \mathrm{SO_4^{2-}}\,, \tag{4}$$

$$>\mathrm{CO_3H^0} \rightleftharpoons >\mathrm{CO_3^-} + \mathrm{H^+}\,, \tag{5}$$

$$>\mathrm{CO_3Ca^+} \rightleftharpoons >\mathrm{CO_3^-} + \mathrm{Ca^{2+}}\,, \tag{6}$$

$$>\mathrm{CO_3Mg^+} \rightleftharpoons >\mathrm{CO_3^-} + \mathrm{Mg^{2+}}\,. \tag{7}$$

The ">"-symbol is used to distinguish surface complexes from aqueous complexes; note that $>CaH_2O^+$ and $>CO_3^-$ are treated as basis species, while the rest are secondary species. All of the surface complex reactions conserve charge. For each reaction there is a corresponding law of mass action expression. Mathematically, the only difference compared to the usual dissociation reactions is the inclusion of a new term that contains the surface potential, to correct activities at the surface for electrostatic effects [10]. As an example, applying the law of mass action to equation (1) yields:

$$
\begin{aligned}
>CaCO_3^- &\rightleftharpoons >CaH_2O^+ + HCO_3^- - H^+ - H_2O, \\
K &= \frac{[>CaH_2O^+]\, a_{HCO_3^-}\, \exp\{F\,\psi/(R\,T)\}}{[>CaCO_3^-]\, a_{H^+}\, \exp\{-F\,\psi/(R\,T)\}}\,, \\
\log_{10} K &= \log_{10}[>CaH_2O^+] \log_{10} a_{HCO_3^-} - \log_{10} a_{H^+} - \log_{10}[>CaCO_3^-] \\
&\quad + 2\,\underbrace{\frac{F\psi_0}{\ln 10\, R\,T}}_{\log_{10} E_0}\,. \qquad (8)
\end{aligned}
$$

The activity of water has been assumed to be unity. To account for electrostatic effects, the activity of an aqueous species that reacts with a surface complex is influenced by the surface potential via the Boltzmann factor,

$$\exp\left(\frac{-Z_i F \psi_0}{RT}\right)\,,$$

where Z_i is the valence of the aqueous species, $F = 9.648456 \cdot 10^4$ C/mol is Faraday's constant, $R = 8.314$ J/(K mol) is the ideal gas constant, ψ_0 (V) is the surface potential, and T is the absolute temperature. Thus, when the surface potential has a negative sign, positively charged ions will have a high activity close to the surface, while negatively charged ions will have a high activity close to the surface if the surface potential is positive.

To accommodate the electrostatic effect into the governing equations of the geochemical model, we have introduced E_0 as a shorthand for the Boltzmann correction. This is not merely a notational convenience: by adding E_0 as added as a fictive basis species, along with the two new surface basis species $>CaH_2O^+$ and $>CO_3^-$, we can write the reaction in exactly the same mathematical form as chemical reactions in the aqueous phase [1]. More generally, we let $E = E(x)$ denote the Boltzmann factor

$$\exp\left(\frac{-Z_i F \psi(x)}{RT}\right)\,,$$

that is, E depends on the potential at distance x from the surface.

2 Example: Equilibrium between solution and carbonate surface site

Terminal

```
geochem
equilibrate 0
solution 0
--Solution to equilibrate
pH 7 charge
Na 0.48
SO4 0.24
/end

complex
method 1
#specific surface area m^2/L pore volume, size of diffusion layer
s_area 1000.0
#Name sites/nm^2
GCa 2
GCO3 2
/end

/end
```

Explanation of input file: Note that the solution keyword is described in the solution example. To model surface interactions with a charged surface of e.g. a carbonate, we need to invoke the `COMPLEX` keyword.

'geochem .. /end': Defines the geochemical input block

'equilibrate': equilibrate without changing the bulk solution (see also the ion exchange example)

'COMPLEX .. /end': Defines an ion exchange calculation block.

'method': can take on three different values, described below

- 0: puts surface potential to zero
- 1: uses the trick described in [4], where one assumes that the solution does not contain ions with higher valence than 2. (Note that many other solvers assumes monovalent solutions)
- 2: uses the general formulation of the Grahame equation (see e.g. [10])

'GCa': name of the (positive) surface site

‘GCO3‘: name of the (negative) surface site

‘2‘: is the number of surface sites, for carbonate typical value is 2-5.

When run, `GeoChemX` will:

1. Create a solution at pH 7.
2. Add the dissolved ions Na^+, SO_4^{2-}.
3. Perform a charge-balance adjustment using H^+.
4. Add surface complexes 2 sites/nm.
5. Compute how positive and negative ions, according to equation (1)-(7), partition between:
   - aqueous phase
   - surface complex sites

2.1 Description of output

In the file `<root_name>_solution.out` there are now the following output

‘pH = 10.6517‘: Final equilibrium pH, the solution becomes alkaline due to surface reactions and sulfate interactions.

‘Sch [C/m2 = –0.0747877‘:] Total surface charge density. The surface is negatively charged, dominated by deprotonated surface carbonate groups.

‘Sch DL= 0‘: Charge in the diffuse layer, zero here because it is not calculated (see next section).

‘psi [mV = –43.5162‘:] Electrostatic potential. The surface develops a negative electric potential consistent with negative charge.

‘F = 96484.6 C/mol‘: Faraday constant. Provided for completeness.

‘S = 1000 m2/L‘: Surface area per liter

‘cbal = –1.01E-7 eq/L‘: Charge balance error. Very small, i.e. good numerical charge balance.

‘n(1), n(-1), n(2), n(-2)‘: Concentration of charge of ions with valence 1, -1, *larger than* 1, and *less than -1*, respectively. These are only calculated when surface model is used.

‘g(1), g(-1), g(2), g(-2)‘: Ratio of surface excess to bulk concentration for valence 1, -1, *larger than* 1, and *less than -1* species, respectively. All zero as the diffusive layer is not calculated (see next section).

References

[1] C. Anthony J. Appelo and Dieke Postma. *Geochemistry, Groundwater and Pollution*. CRC Press, 2nd edition edition, 2005.

[2] Philippe Van Cappellen, Laurent Charlet, Werner Stumm, and Paul Wersin. A surface complexation model of the carbonate mineral-aqueous solution interface. *Geochimica et Cosmochimica Acta*, 57(15):3505–3518, 1993.

[3] A. Hiorth, L. Cathles, and M. Madland. The impact of pore water chemistry on carbonate surface charge and oil wettability. *Transport in Porous Media*, 85:1–21, 2010.
url 10.1007/s11242-010-9543-6.

[4] Oddbjorn Nodland and Aksel Hiorth. A new formulation of the surface charge/surface potential relationship in electrolytes with valence less than three. *Computational Geosciences*, 28(2):289–304, 2024.

[5] Oleg S. Pokrovsky and Jacques Schott. Processes at the magnesium-bearing carbonates/solution interface. ii. kinetics and mechanism of magnesite dissolution. *Geochimica et Cosmochimica Acta*, 63(6):881–897, 1999.

[6] Oleg S. Pokrovsky and Jacques Schott. Kinetics and mechanism of dolomite dissolution in neutral to alkaline solutions revisited. *American Journal of Science*, 301(7):597–626, 2001.

[7] Oleg S. Pokrovsky, Jacques Schott, and Fabien Thomas. Dolomite surface speciation and reactivity in aquatic systems. *Geochimica et Cosmochimica Acta*, 63(19-20):3133–3143, 1999.

[8] Oleg S. Pokrovsky, Jacques Schott, and Fabien Thomas. Processes at the magnesium-bearing carbonates/solution interface. i. a surface speciation model for magnesite. *Geochimica et Cosmochimica Acta*, 63(6):863–880, 1999.

[9] OS Pokrovsky and J. Schott. Surface chemistry and dissolution kinetics of divalent metal carbonates. *Environmental science & technology*, 36(3):426–432, 2002.

[10] W. Stumm, L. Sigg, and B. Sulzberger. *Chemistry of the Solid-Water Interface: Processes at the Mineral-Water and Particle-Water Interface in Natural Systems*. A Wiley-interscience publication. Wiley, 1992.